

IBRAHIM CHEHAB

MECHANICAL ENGINEERING STUDENT

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Dearborn Heights, Mi

ABOUT ME

Motivated Mechanical Engineering student dedicated to achieving meaningful outcomes in vehicle engineering systems design, development, and testing. Enthusiastic about automotive culture and embracing dynamic challenges through hands-on mechanical experience. Proven expertise in 3D CAD modeling, technical problem-solving, and effective collaboration on engineering project teams. Proficient in employing rapid prototyping tools and precise mathematical modeling to drive innovative product development.

EDUCATION

Bachelor of Science in Mechanical Engineering | Expected Graduation: May 2029

University of Michigan-Dearborn

- Standing: Junior
- Relevant Coursework: Engineering Computations, Fluid Mechanics, Thermodynamics, SolidWorks and Fusion

PROFESSIONAL & PROJECT EXPERIENCE

Formula SAE Aerodynamics Team Member | University of Michigan-Dearborn | Sept 2025 – Present

- Designed and optimized race vehicle aerodynamics to increase downforce, directly applying vehicle engineering systems design principles.
- Utilized computational fluid dynamics software and physical wind tunnel models to validate engineering concepts and improve overall vehicle performance.
- Collaborated within a fast-paced mechanical project team environment to meet strict manufacturing and testing deadlines.

Mechanical Design & Prototyping Projects | Academic & Independent | Aug 2025 – Present

- Created comprehensive 3D CAD models and technical engineering drawings for mechanical components including polygon axles, slotted spring pins, and bolt circles, using SolidWorks.
- Operated a Bambu Lab P1S 3D printer with an AMS system to rapidly prototype parts and fabricate custom mechanical adapters for hardware integration.
- Developed advanced MATLAB and Python scripts for precise mathematical calculations, symbolic differential equation solving, and 3D trajectory visualization.

Group Leader & Specials Teacher | Wayne Metro & Hype Athletics | Jun 2025 – Present

- Managed structured programs and coordinated group dynamics for large groups of students, demonstrating strong organizational and leadership skills.
- Maintained high accountability, adaptability, and reliable attendance, establishing the work ethic necessary to commit to a 40-hour work week.

TECHNICAL SKILLS

SolidWorks

3D Printing

MS Office Suite

MATLAB

Rapid Prototyping

Data Analysis

Python

Aerodynamic Testing

Mechanical Troubleshooting

Blender

Mathematical Modeling

ADDITIONAL INFORMATION

- **Languages:** English, Arabic.
- **Interests:** Automotive Culture, Formula 1 Technical Regulations, Race Strategy Analysis.
- **Leadership & Volunteer:** High School Varsity Golf Captain, National Honor Society (227+ hours of community service).